Exam Presentation

Probability Theory, Stochastic Processes and Applied Statistics

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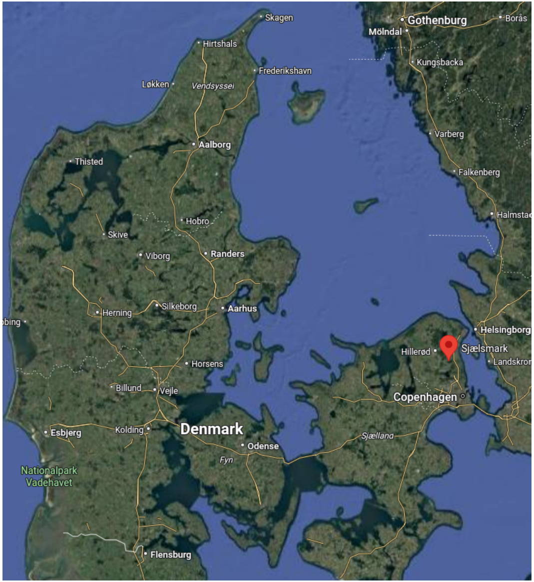
2026

Workshop 1: Wind Speed Distributions

Motivation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Flensburg

The Weibull Distribution

A random variable X follows a Weibull distribution with parameters $k > 0$ (shape) and $\lambda > 0$ (scale):

$$X \sim \text{Weibull}(k, \lambda)$$

$$f(x) = \begin{cases} \frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

- k controls the shape
- λ stretches the distribution

Distribution Function

The distribution function is

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(x)dx$$

For the Weibull distribution:

$$F(x) = 1 - e^{-(x/\lambda)^k}, \quad x > 0$$

- Obtained by integrating the density using the substitution $u = (x/\lambda)^k$

We can also prove:

$$\ln\big(-\ln(1 - F(x))\big) = -k\ln(\lambda) + k\ln(x)$$

Mean and Variance

The mean of a Weibull distributed variable is

$$\mu = \int_{-\infty}^{\infty} x \cdot F(x) dx$$

$$\mu = \lambda \Gamma\left(1 + \frac{1}{k}\right)$$

The variance is

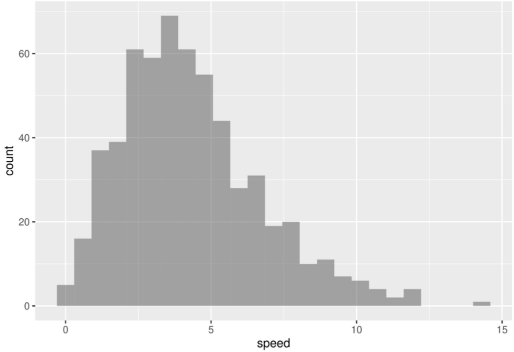
$$\mathrm{Var}(X) = \int_{-\infty}^{\infty} x^2 \cdot F(x) dx - \mu^2$$

$$\mathrm{Var}(X) = \lambda^2 \left(\Gamma\left(1 + \frac{2}{k}\right) - \Gamma\left(1 + \frac{1}{k}\right)^2\right)$$

$$\sigma = \lambda \sqrt{\Gamma\left(1 + \frac{2}{k}\right) - \Gamma\left(1 + \frac{1}{k}\right)^2}$$

Wind Speed Data

First step: visualize the data using a histogram



- Wind speed is:
  - Positive
  - Right-skewed
  - Not normally distributed

Empirical Distribution Function

Let the ordered observations be

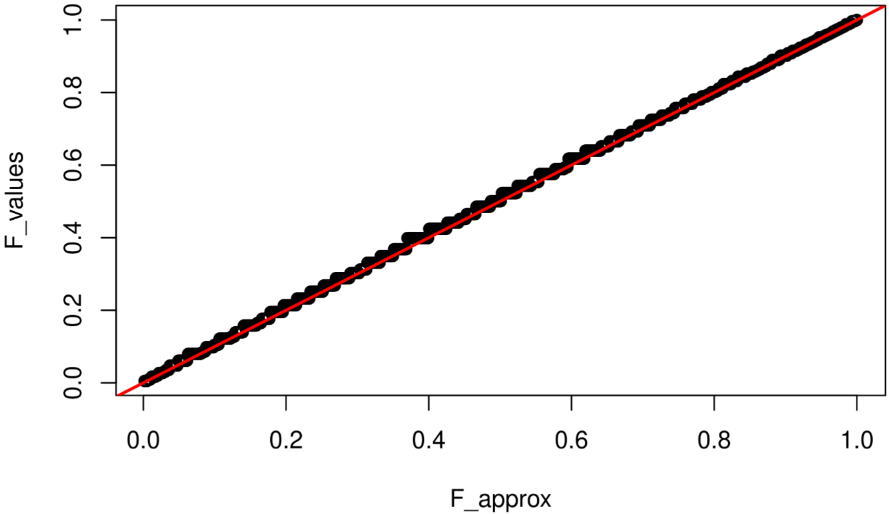
$$x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$$

For the ith ordered value:

$$F(x_{(i)}) \approx \frac{i}{n}$$

- Exactly i observations are $\leq x_{(i)}$
- This approximates the true CDF

Weibull Probability Plot



Weibull Probability Plot

Define

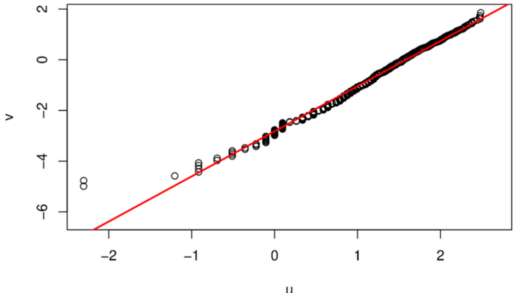
$$u_i = \ln(x_{(i)}), \quad v_i = \ln\left(-\ln\left(1 - \frac{i}{n}\right)\right)$$

Remember

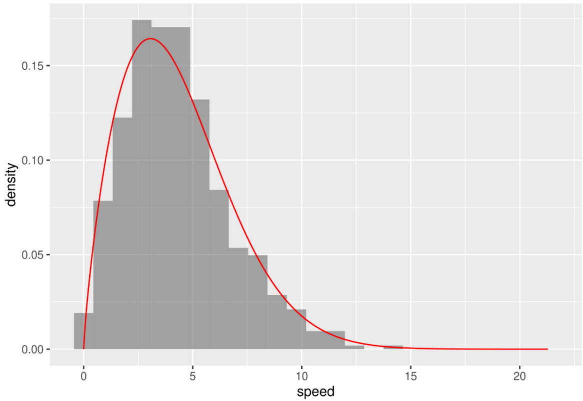
$$\ln\left(-\ln(1 - F(x))\right) = -k\ln(\lambda) + k\ln(x)$$

If data follows a Weibull distribution:

$$v_i \approx -k\ln(\lambda) + ku_i$$



Weibull Probability Plot



k and λ are calculated from the intercept and slope:

- $k = 1.78$
- $\lambda = 4.87$

Sample Moments and CLT

Let $X_1, \ldots, X_n$ be i.i.d. Weibull variables and

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

By the Central Limit Theorem:

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

for sufficiently large n.

Sample Moments and CLT

From before we know:

$$\mu = \lambda \Gamma\left(1 + \frac{1}{k}\right)$$

$$\sigma = \lambda \sqrt{\Gamma\left(1 + \frac{2}{k}\right) - \Gamma\left(1 + \frac{1}{k}\right)^2}$$

Evaluating with $k = 1.78$ and $\lambda = 4.87$,

$$\mu = 4.333 \text{ and } \sigma = 3.766$$

Simulation Study

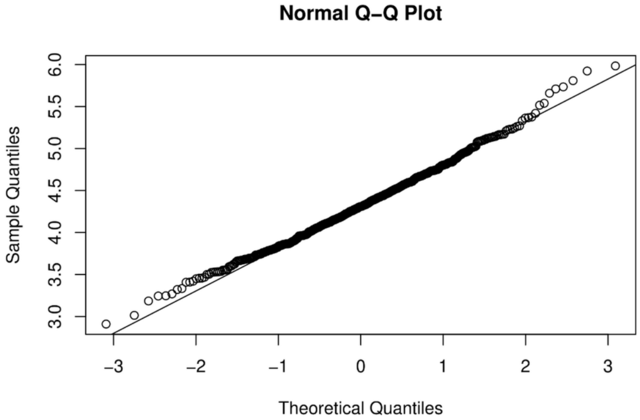
- Sample size: $n = 30$
- Repeat sampling 500 times
- Store each sample mean

We examine:

- Mean of sample means
- Standard deviation (standard error)
- Shape of the distribution

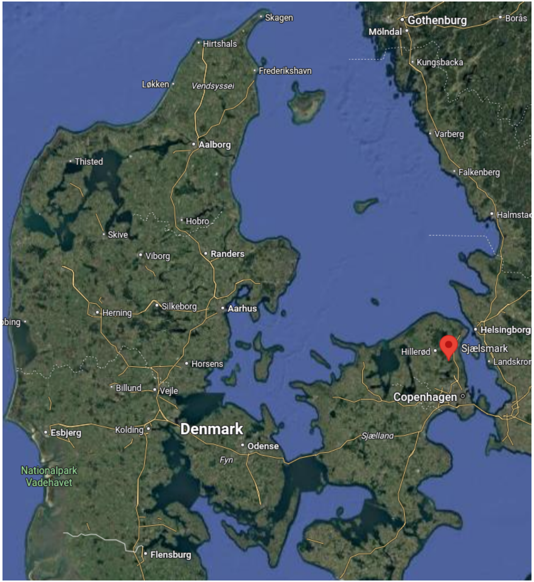
Results

- Mean of $\bar{X}$ is close to μ ($\bar{x} = 4.326$, $\mu = 4.333$)
- Standard deviation is close to $\sigma/\sqrt{n}$ ($SD = 0.490$, $\sigma/\sqrt{n} = 0.688$)
- QQ-plot shows approximate normality



Workshop 2: Seasonal Wind Speed

Motivation



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Seasonal Wind Speed Data

Quarterly average wind speeds (2001–2019) at Sjælsmark, Denmark:

quarters: spring, summer, autumn, winter

- Assumption: Quarterly averages $\approx$ Normal (CLT)
- Significance level: $\alpha = 0.05$

Comparison of Spring and Autumn: Variances

Null hypothesis:

$$H_0 : \sigma^2_{\text{spring}} = \sigma^2_{\text{autumn}}, \quad H_A : \sigma^2_{\text{spring}} \neq \sigma^2_{\text{autumn}}$$

- F-test statistic: $F = s^2_{\text{spring}} / s^2_{\text{autumn}}$
- Degrees of freedom: $df_{\text{spring}} = n_{\text{spring}} - 1$, $df_{\text{autumn}} = n_{\text{autumn}} - 1 \Rightarrow df = 18$
- Two-sided p-value: 0.704 $\Rightarrow$ do not reject H_0 $\Rightarrow$ We can assume the variances in spring and autumn are approximately equal.

Comparison of Spring and Autumn: Means

Null hypothesis:

$$H_0 : \mu_{\text{spring}} = \mu_{\text{autumn}}, \quad H_A : \mu_{\text{spring}} \neq \mu_{\text{autumn}}$$

- Pooled standard deviation:

$$s_p = \sqrt{\frac{(n_{\text{spring}} - 1)s^2_{\text{spring}} + (n_{\text{autumn}} - 1)s^2_{\text{autumn}}}{n_{\text{spring}} + n_{\text{autumn}} - 2}}$$

- Standard error: $SE = s_p \sqrt{\frac{1}{n_{\text{spring}}} + \frac{1}{n_{\text{autumn}}}}$
- t-statistic: $t = (\bar{x}_{\text{spring}} - \bar{x}_{\text{autumn}})/SE$
- p-value: $0.0059 < 0.05 \Rightarrow$ reject $H_0 \Rightarrow$ The difference in means is statistically significant.

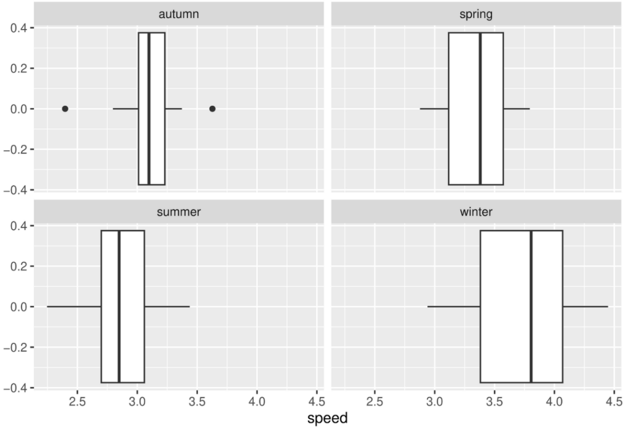
Comparison of Spring and Autumn: Means

$$CI = (\bar{x}_{\text{spring}} - \bar{x}_{\text{autumn}}) \pm t_{1-\alpha/2,df} \cdot SE$$

$$CI = [0.077, 0.427]$$

- Spring winds significantly higher than autumn
- Interpretation: 95% confident that the difference in means is positive

Boxplots of Quarterly Wind Speeds



- Box: Q1 to Q3, median line at Q2
- Whiskers: $[Q1 - 1.5 \cdot IQR, Q3 + 1.5 \cdot IQR]$
- Outliers shown as dots
- Winter shows highest median, summer the lowest

Testing Differences Across All Quarters

Null hypothesis:

$$H_0 : \mu_{\text{spring}} = \mu_{\text{summer}} = \mu_{\text{autumn}} = \mu_{\text{winter}}$$

- Model: *speed* $\sim$ *quarter*
- F-test: p-value $< 0.001 \Rightarrow$ reject $H_0 \Rightarrow$ the difference in means is statistically significant.
- Multiple R-squared: $0.52 \rightarrow 52\%$ of total variability explained by quarter

Tukey's Test

Tukey test adjusts for multiple testing:

- Winter $>$ all other quarters (all p-values $\ll 0.05$)
- Summer vs spring: spring significantly higher
- Spring-autumn and summer-autumn: not significant after adjustment
- CI for spring-autumn difference (Tukey): $[-0.012, 0.516]$
  - Wider than t-test CI, includes 0
  - Multiple comparison adjustment reduces significance

```
##
## Fit: aov(formula = speed ~ quarter, data = wind)
##
## $quarter
##                     diff         lwr         upr     p adj
## spring-autumn  0.2521376 -0.01211513  0.51639024 0.0668933
## summer-autumn -0.2153825 -0.47963516  0.04887022 0.1492955
## winter-autumn  0.6272860  0.36303327  0.89153864 0.0000002
## summer-spring -0.4675200 -0.73177271 -0.20326734 0.0000843
## winter-spring  0.3751484  0.11089571  0.63940109 0.0020779
## winter-summer  0.8426684  0.57841573  1.10692111 0.0000000
```

Workshop 3: Battery Capacity

Battery Capacity Data

We study battery capacity loss Q measured at different temperatures.

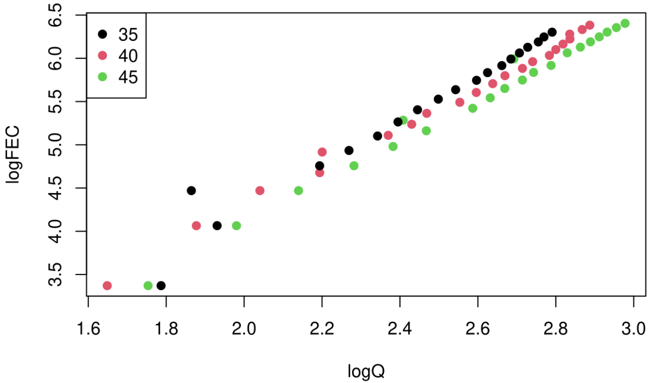
- Response variable: capacity loss Q (in %)
- Predictors:
  - Number of charge cycles (FEC)
  - Temperature: 35°, 40°, 45° C
- Log-transformed variables:

$$\log(Q), \quad \log(\text{FEC})$$

- Temperature treated as a categorical variable

Exploratory Analysis

Scatter plot of log(Q) versus log(FEC) colored by temperature.



- Strong linear relationship
- Higher temperatures shift the relationship

Correlation Analysis

Correlation between $\log(Q)$ and $\log(\mathrm{FEC})$ for each temperature:

$$r_{35} = 0.98, \quad r_{40} = 0.997, \quad r_{45} = 0.994$$

- Correlation measures strength of linear dependence
- Values close to 1 indicate very strong linear association
- Confirms visual impression from the scatter plot

Multiple Regression Model (No Interaction)

We consider the model:

$$\log(Q_i) = \beta_0 + \beta_1 \log(\mathsf{FEC}_i) + \beta_2 \mathbb{I}(T_i = 40) + \beta_3 \mathbb{I}(T_i = 45) + \varepsilon$$

- α: intercept
- β_1: effect of charge cycles at 35C
- β_2: effect of 40C
- β_3: effect of 45C
- $\varepsilon \sim N(0, \sigma^2)$

Estimated Regression Model

Estimated model:

$$\log(Q_i) = 0.224846 + 0.411014\log(\mathrm{FEC}_i) + 0.043310\mathbb{I}(T_i = 40) + 0.106577\mathbb{I}(T_i = 45) + \varepsilon_i$$

- log(FEC) has a strong positive effect
- Higher temperatures increase capacity loss
- Predictors are statistically significant

Inference and Model Fit

- t-statistics computed as:

$$t = \frac{\text{Estimate}}{\text{Std. Error}}$$

- The amount of std. errors away each coefficient is from 0
- Multiple $R^2 = 0.982$
- Interpretation:
- 98% of the variability in $\log(Q)$ is explained

Overall F-Test

Null hypothesis:

$$H_0 : \beta_1 = \beta_2 = 0$$

- F-test compares explained to unexplained variance
- p-value < 0.001
- Reject H_0
- At least one predictor significantly affects $\log(Q)$

Interaction Model

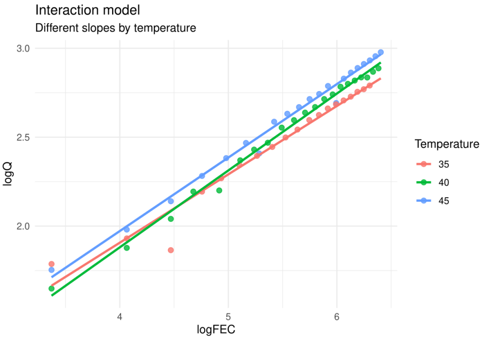
We test whether temperature modifies the effect of charge cycles:

$$\log(Q_i) = \beta_0 + \beta_1 \log(\text{FEC}_i) + \beta_2 \mathbb{I}(T_i = 40) + \beta_3 \mathbb{I}(T_i = 45)$$

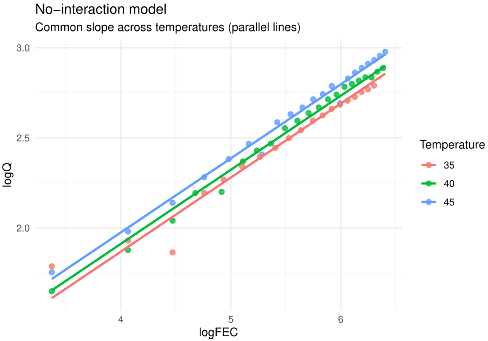
$$+\beta_4 \log(\text{FEC}_i)\mathbb{I}(T_i = 40) + \beta_5 \log(\text{FEC}_i)\mathbb{I}(T_i = 45) + \varepsilon_i$$

- Compare models with and without interaction
- F-test: $p = 0.13$
- No significant interaction

Model Fits

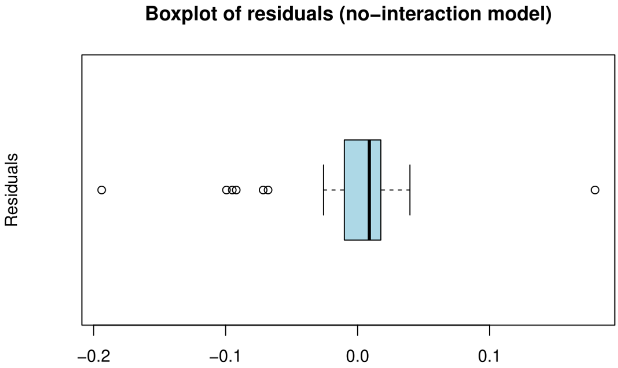


Interaction model
Different slopes by temperature
3.0
2.5
2.0
logQ
4
5
6
logFEC
Temperature
35
40
45



No−interaction model
Common slope across temperatures (parallel lines)
3.0
2.5
2.0
logQ
4
5
6
logFEC
Temperature
35
40
45

Residual Analysis



- Median residual close to zero
- Indicates unbiased predictions
- Some outliers present

Conclusions

- Battery capacity loss increases with:
  - Number of charge cycles
  - Temperature
- Linear regression explains most variability
- No evidence of interaction between predictors

Workshop 4: DC Motor Model

Workshop 1: Wind Speed Distributions

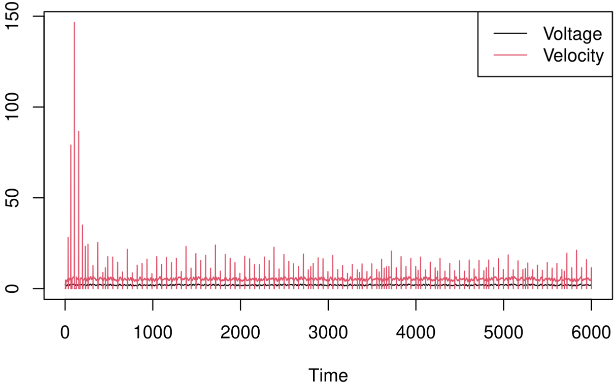
Workshop 2: Seasonal Wind Speed

Workshop 3: Battery Capacity

Workshop 4: DC Motor Model

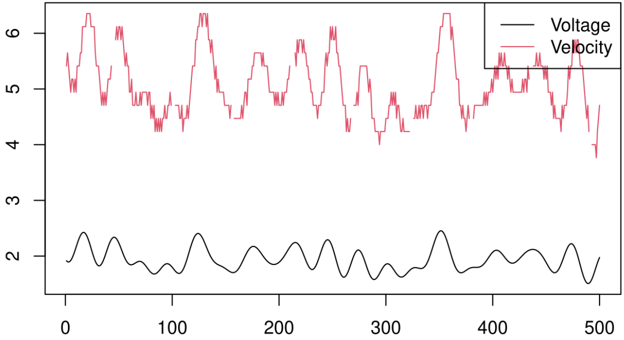
Problem: DC Motor

- Input-output time series
- Noise and disturbances
- Goal: Model motor velocity as a function of input voltage



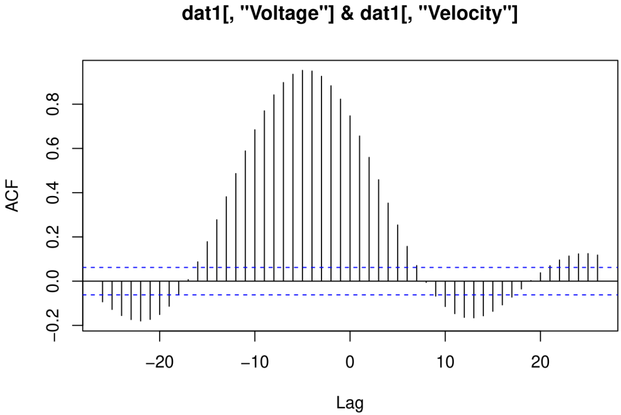
Data Visualization and Cleaning

- Raw data: voltage (V) and motor velocity (rad/s)
- Sampling time: 0.01 s
- Zoom-in reveals noise and periodic disturbances
- Data cleaning:
  - Remove first 500 samples
  - Set velocities outside $[3, 7]$ rad/s to NA
  - Pick interval $t = 1000 : 2000$ for analysis



Cross-Correlation Function (CCF)

- Detect delay between input voltage and output velocity
- CCF peak indicates system delay
- Maximum correlation → estimated lag: -5 samples
- Sampling time = 0.01 s → delay = -0.05 s



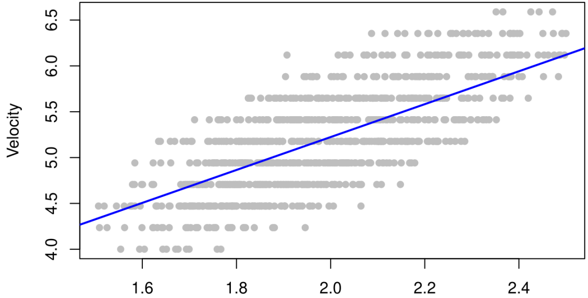
Linear Regression with AR(1) Noise

$$\omega_t = \beta_0 + \beta_1 V_t + \alpha\epsilon_{t-1} + W_t$$

- Fit AR(1) model with input voltage as exogenous variable
- Estimated parameters:

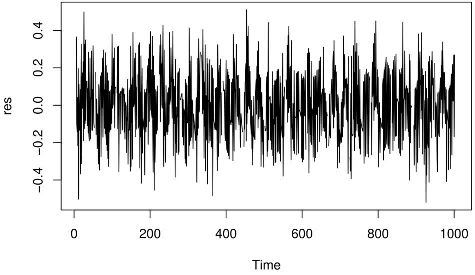
$$\beta_0 = 1.6348, \quad \beta_1 = 1.7945, \quad \alpha = 0.8584$$

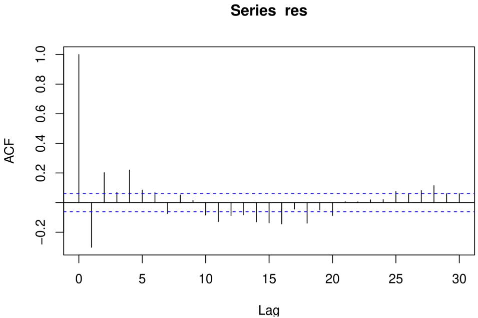
- Noise ACF: $\rho(h) = \alpha^h = 0.8584^h$



Residual Analysis for AR(1) Model

- Residuals should resemble white noise
- Check time series plot and correlogram
- Some oscillations remain from original periodic disturbance



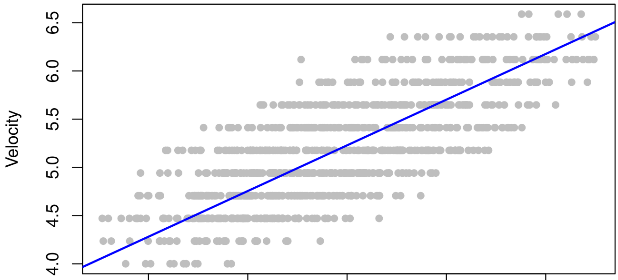


ARMA(p,q) Model Selection

- Fit ARMA(p,q) models with $p, q = 0, 1, 2, 3$
- Compare using AIC
- Best model: ARMA(3,3) with lowest AIC

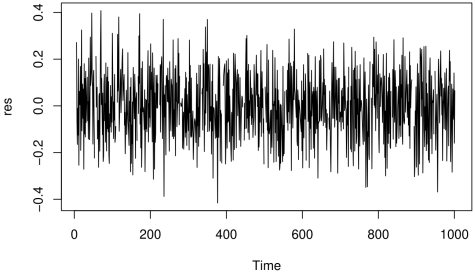
$$\omega_t = \beta_0 + \beta_1 V_t + n_t, \; n_t = \sum_{i=1}^{3} \phi_i n_{t-i} + \sum_{j=1}^{3} \theta_j \epsilon_{t-j} + \epsilon_t$$

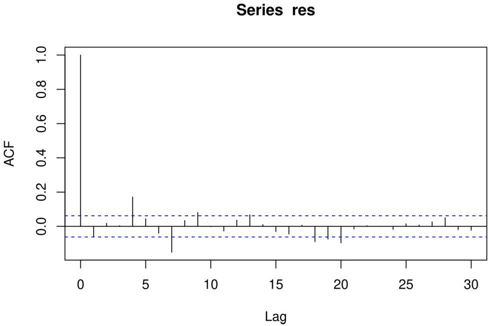
$$\omega_t = 0.4916 + 2.3687 V_t + n_t$$



ARMA(3,3) Residuals

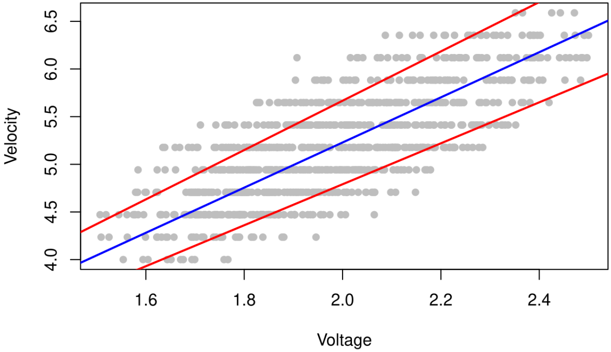
- Residuals checked for whiteness
- Correlogram confirms good fit





Regression Slope and Confidence Interval

- Estimated slope: $\beta_1 = 2.3687$
- 95% CI: $[2.145, 2.592]$
- Graphical visualization:



Prediction Example

- Next voltage: $V = 2.23$
- Predicted velocity: $\hat{\omega} = 5.8535$ rad/s
- 95% prediction interval: $[5.539, 6.168]$ rad/s